# CSC-580 PM CASE: THIRD AVENUE SOFTWARE HEALTH-CARE APP PROJECT

This case is new for the ninth edition of *Information Technology Project Management*. The case provides an opportunity to apply agile and Scrum principles to project management.

Each part of the case contains several task assignments to help you explore the use of agile and Scrum principles.

## Part 4: Project Cost and Quality Management

As with other methods of software development, the application's size is a major indicator of how much it might cost to develop. In agile development, cost estimates are often made based on size measurements such as story points.

You will recall that each user story for the health-care app is assigned several story points based on estimates of how much work is needed to complete each major task. Based on prior experience with agile projects, the three-person accounting staff at Third Avenue Software has determined an average dollar production cost for a story point: $1200.

However, the staff accountants are not completely confident in this average dollar value because Third Avenue's experience with agile projects is not extensive. The accountants would like the team to confirm their calculations, if possible.

As you learned in Module 7, earned value management (EVM) is a more traditional project management method for determining whether a project is meeting time and cost goals. EVM requires calculation of three values for each major activity in the project:

- The planned value (PV), which is the authorized budget assigned to scheduled work.
- The actual cost (AC), which is the realized cost for the work performed on an activity during a specified timeframe.
- The earned value (EV), a measurement of project work done, framed in terms of the approved budget for the work.

Several more values and formulas are involved in EVM. For more information, refer to the "Earned Value Management" section of the Module 7 Reading.

Managers at Third Avenue are also eager to see evidence that the Quality Assurance staff are making progress in their ability to test the health-care app. They have asked the team to provide at least a basic framework of test specifications.

*• Open a new Microsoft® Word document and complete the **Tasks** below. • Save the file on your computer with your last name in the file name. (Example: part 1 tasks _Jones.doc)*
*• Click the **Choose File** button to select and upload your saved document.*

***Tasks***

1. Review the user stories, technical stories, and story points you developed in earlier parts of this case and then use the information to estimate the cost of the health-care app project. As you learned in the introduction to Part 4, Third Avenue accountants estimate that the average production cost of a story point is $1200. Based on the values you calculate, do the project costs appear to be within the overall budget of $350,000? Do the costs seem to be within the MVP budget of $120,000? If the costs are not within budget, adjust the story points you developed in Part 3 to make sure your revised numbers are within budget. You might also have to reconsider the feature set within the MVP if it exceeds its budget.

Use Case Name: Payment Tracker Story Points: 9

Use Case Name: Web Resources Story Points: 2

Use Case Name: Emergency info Story Points: 3

Use Case Name: Emergency List Story Points: 10

Use Case Name: Contact Data Story Points: 5

Use Case Name: Medication Tracker Story Points: 8

Use Case Name: Fitness Tracker Story Points: 3

MVP Deliverables: Contact Data, Emergency List

Story points: 15

Cost @ $1200/SP: $18,000

Adjusted multiplier for realism: 6

Revised Story Points:

Use Case Name: Payment Tracker Story Points: 54

Use Case Name: Web Resources Story Points: 12

Use Case Name: Emergency info Story Points: 18

Use Case Name: Emergency List Story Points: 60

Use Case Name: Contact Data Story Points: 30

Use Case Name: Medication Tracker Story Points: 48

Use Case Name: Fitness Tracker Story Points: 18

Revised Realistic Story Points: 90

Revised Realistic MVP cost: $108,000

Total Story Points: 240

Total Cost: $288,000

Even after adjusting the story point values to more accurately represent the real life scenario and budget estimates given the estimate of $1,200/SP it looks as if the Total project will fall well within budget. Preliminary estimates put the overall project cost at $288,000 which is $62,000 favorable to the $350,000 budget.

During the original scope formulation of the MVP deliverable discussed in Question 4 of Part 2 of the Project Management Full case it was decided that the release would have to include both the contact data feature and the emergency list feature. After adjust story point values to a meet a much more realistic estimate to budget we found that developing the product within budget is very possible. Currently it stands that the MVP will be achieved $12,000 favorable to budget given an $108,000 estimate to a budget of $120,000.

2. Assume that you have completed one month of the health-care app project and have some reliable EVM data for cost accounting. Remember that the budget at completion (BAC) is $350,000 for the four-month project. You have received the following figures from the Third Avenue accountants:

   PV $105,000

   EV $122,000

   AC $105,000

   Using this information, answer the following questions.

   • What is the cost variance, schedule variance, cost performance index (CPI), and schedule performance index (SPI) for the project?
   • Use the CPI to determine the estimate at completion (EAC) for the project. • Based on your answers, does the project appear to be on schedule and within budget?

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|---|---|---|---|---|---|---|---|
| PV | $105,000 | | CV=EV-AC | SV=EV-PV | SPI=EV/PV | CPI=EV/AC | EAC=BAC/CPI |
| EV | $122,000 | | $17,000 | $17,000 | 116.19% | 116.19% | $301,229.51 |
| AC | $105,000 | | | | | | |
| BAC | $350,000 | | | | | | |

Given an SPI and CPI of ~%116 not only do we see that the project is on time and on budget, we are actually ahead of schedule. Currently we can expect to achieve the project with an overall reduction of %16 percent of the total time estimated to complete. Also, our EAC of $301,229.51 shows us that we are on track to come in ~$49,000 favorable to our budget of $350,000

3. The health-care app will undergo rigorous software testing by the Third Avenue Quality Assurance staff, using test plans that might fill entire binders. At this early point in the process, however, such plans are still in development. Develop a short list of quality requirements for testing at least five of the important app features and/or usability issues described thus far in this running case. In your list, briefly describe each requirement.

   For example, here is a short (and far from complete) list of quality requirements for the emergencies list feature, which stores vital phone numbers and addresses as well as provides quick access to local in-network hospitals, urgent care clinics, and children or friends who can be relied upon to provide transportation in an emergency:

   - Do all the data-entry fields accept text without problems? For example, long entries in a field can sometimes be clipped or truncated prematurely, which is frustrating to users.
   - Is it possible to enter nonsensical data in a field? For example, users should not be allowed to type text characters into a field that records phone numbers.
   - Does the software automatically move the user from one entry field to the next? Does the cursor always appear in the next entry field as a means of visual feedback for the user?
   - As soon as the user enters and saves an address, an interactive GPS map should become available in a new window, with voice and text directions. Does the window appear? Does it appear immediately?
   - When you slide your finger to create the effect of movement within the map, does the software redraw in a timely manner?

Emergency List:

- Do all the data-entry fields accept text without problems? For example, long entries in a field

can sometimes be clipped or truncated prematurely, which is frustrating to users.

- Is it possible to enter nonsensical data in a field? For example, users should not be allowed to type text characters into a field that records phone numbers.

- Does the software automatically move the user from one entry field to the next? Does the cursor always appear in the next entry field as a means of visual feedback for the user?

- As soon as the user enters and saves an address, an interactive GPS map should become available in a new window, with voice and text directions. Does the window appear? Does it appear immediately?

- When you slide your finger to create the effect of movement within the map, does the software redraw in a timely manner?

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Payment Tracker:

· If the user picks the link to take them to the payment tracker option does it actually take them to the designated place?

· Does the feature bring up a reasonable amount of transactions? Ideally they app would show the first few transactions and allow the user to view more history if necessary.

· Does the feature track only the expenses that are health related? The feature should not show purchases the aren't relevant to Health Expenses.

· Does the feature securely link the users bank account information without exposing their personal account information directly?

Web resources:

· If the user picks the link to take them to the Web resources option does it actually take them to the designated place?

· Do the websites rotate regularly to avoid stagnant content? Ideally the websites are related to health information found elsewhere in the app and refresh on a regular basis to avoid being stagnant.

· If the user selects a given article or web resource does it take them to the place the link described? Ideally it doesn't 404 or aggregate things that don't exist anymore.

Emergency info:

- If the user picks the link to take them to the Emergnecy info option does it actually take them to the designated place?

- Is it possible to enter nonsensical data in a field? For example, users should not be allowed to type text characters into a field that records phone numbers.

- Are the database links set up correctly? If the user gives information is it stored in a way that the program can relate, display, and utilize it effectively?

- After entering information does the screen refresh accordingly? The feature should appear dynamic instead of static.

- If the User updates some information is the database updated accordingly? Things such as case sensitive areas and such should be dynamic enough to not create new entries with slightly different inputs.

Contact data:

- If the user picks the link to take them to the contact data option does it actually take them to the designated place?

- Is it possible to enter nonsensical data in a field? For example, users should not be allowed to type text characters into a field that records phone numbers.

- Are the database links set up correctly? If the user gives information is it stored in a way that the program can relate, display, and utilize it effectively?

- After entering information does the screen refresh accordingly? The feature should appear dynamic instead of static.

- If the User updates some information is the database updated accordingly? Things such as case sensitive areas and such should be dynamic enough to not create new entries with slightly different inputs.

- If the user selects a method of contact does it relay the correct method? If I choose email does it email or does it try to bring up a phone call?

Medication Tracker:

- If the user picks the link to take them to the Medication Trackeroption does it actually take

them to the designated place?

- Is it possible to enter nonsensical data in a field? For example, users should not be allowed to type text characters into a field that records phone numbers.

- Are the database links set up correctly? If the user gives information is it stored in a way that the program can relate, display, and utilize it effectively?

- After entering information does the screen refresh accordingly? The feature should appear dynamic instead of static.

- If the User updates some information is the database updated accordingly? Things such as case sensitive areas and such should be dynamic enough to not create new entries with slightly different inputs.

- Does the calendar operate appropriately? Sometimes it is a leap year and may have less days in February for example. The calendar must be updated every year.

- If given a prescribed schedule does the feature send notifications when it is supposed to? Ambiguous selections should not be possible and user should have to put in information that can be related to OS's way of carrying out tasks.

- If a prescribed frequency is input does the calendar update accordingly?

Fitness Tracker:

- If the user picks the link to take them to the Fitness Tracker option does it actually take them to the designated place?

- Is it possible to enter nonsensical data in a field? For example, users should not be allowed to type text characters into a field that records phone numbers.

- Are the database links set up correctly? If the user gives information is it stored in a way that the program can relate, display, and utilize it effectively?

- After entering information does the screen refresh accordingly? The feature should appear dynamic instead of static.

- If the User updates some information is the database updated accordingly? Things such as case sensitive areas and such should be dynamic enough to not create new entries with slightly different inputs.

- Does the app display information from the database in a clear, concise and engaging manner

to the user?

· If the user has a wearable does the feature appropriately upload, store and display the information that was recorded effectively?

4. Modules 2 and 8 of the text discussed Kanban, a method sometimes used with Scrum. Kanban uses five core properties:

   • *Visualize the workflow*. Cards are a common visualization method.

   • *Limit work in progress*. New project work is done when it can be incorporated into the next software iteration and when there is available capacity. In this way, problem areas are quickly revealed for resolution.

   • *Measure and manage flow*. A key here is to analyze problem areas and then implement changes to correct the problems as quickly as possible.

   • *Make process policies explicit*. Everyone on the project team must understand the processes and any problems with them.

   • *Use models to recognize improvement opportunities*. The models themselves are less important than the important agile concept of continuous improvement.

   Based on your work on the case so far, which property of Kanban has proven most useful to you? Write a two- to three-paragraph answer.

Based on the work so far in the case we have found Kanban's unique and effective ability to Visualize the workflow to be most advantageous. Workflows are directly and indirectly documented throughout the entire project, elements can be found in WBS's, Project Charters, Project Scope Statements, and in multiple other aspects. Although there is ample documentation of workflow throughout the process it readily becomes evident that workflow itself is a very transient and ambiguous concept. If the actual workflow process is not explicitly modeled in a physical representation it opens the opportunity for team members to come to different conclusions on different aspects of the project. In this respect Kanban is highly effective at succinctly modeling the entire workflow process to minimize ambiguity and keep team members in line and on task.

Kanban's visualization of workflow also puts tangible objects to abstract concepts. The advantage can be likened to spending physical money versus purchasing with your debit card. Decisions, deadlines, and work become tangible rather than abstract. When people see how their actions can cue actual physical reactions to a physical model of their project, they can begin to properly gauge time and effort allocated to the project to achieve budget and timing goals.

We can also find parallels of how Kanban visualization can be effective in our current jobs. *Status at a Glance* is often highly sought after by Upper Management when reviewing projects. This can be elusive to projects because there is so much information to display at any given time. Kanban accurately depicts in a quick and concise manner what part of the project you are in, what has been completed, and work still to be done. Management can visually quickly attain status on a quick walk through or review instead of having to designate expensive time for lengthy reviews to display the same information that the board encapsulates in a glance.